

LAB 10 REPORT

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Lab Title: Nested Loop

Theory: A **nested loop** in Python is a loop placed inside another loop.

It is used to perform repeated tasks where one loop controls the execution of another loop. Nested loops are commonly used for patterns, tables, and working with multi-dimensional data.

TASK 1

```
for i in range(1,11):  
    for j in range(1,11):  
        print(i*j,end=" ")  
    print()
```

Output:

```
1 2 3 4 5 6 7 8 9 10  
2 4 6 8 10 12 14 16 18 20  
3 6 9 12 15 18 21 24 27 30  
4 8 12 16 20 24 28 32 36 40  
5 10 15 20 25 30 35 40 45 50  
6 12 18 24 30 36 42 48 54 60  
7 14 21 28 35 42 49 56 63 70  
8 16 24 32 40 48 56 64 72 80  
9 18 27 36 45 54 63 72 81 90  
10 20 30 40 50 60 70 80 90 100
```

Process finished with exit code 0

TASK 2

```
for i in range(1,5):  
    for j in range(5-i):  
        print(" ",end="")  
    for j in range(2*i-1):
```

```
print("\n", end="")  
print()
```

Output:

```
  *  
 ***  
*****  
*****
```

Process finished with exit code 0

TASK 3

```
for i in range(1,6):  
    for j in range(1,i+1):  
        print("\n",end=" ")  
    print()
```

Output:

```
 *  
* *  
* * *  
* * * *  
* * * * *
```

Process finished with exit code 0